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Analysis of energy input-output in direct seeded and transplantation cultivation of rice under different irrigation systems

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ABSTRACT

Energy input-output assessment for crop production processes is becoming increasingly important due to increased energy demand. This study explored the energy consumption for puddled transplanted (PTR), un-puddled transplanted (UPTR), and direct-seeded rice (DSR) cultivation methods under sprinkler, flood and drip irrigation systems in Golestan province, Iran, during 2019 and 2020. The energy of inputs and outputs were calculated using energy equivalents. Then, using input energy values, the global warming potential (GWP) was determined based on the equivalent of carbon dioxide (CO_2). According to the results, the highest amount of yield was obtained for permanent flood irrigation with PTR. The results of mean comparison of cultivation method showed that the highest and lowest amounts of energy input were obtained for PTR (32418 MJ/ha) and UPTR (28175 MJ/ha), respectively. Also, the amounts of indices of energy use efficiency (EUE), energy productivity (EP) and net energy (NE) in DSR were lower than transplanted treatments. The results of mean comparison showed that the lowest amount of GWP was related to UPTR. The results of this study showed that drip irrigation with UPTR cultivation (due to less energy input and a yield close to PTR) can be a technology to save water and energy.

1. Introduction

Due to the energy crisis and the emission of greenhouse gases caused by the excessive consumption of fossil fuels, all efforts are aimed at reducing energy consumption in various sectors industry, transportation, household and commercial as much as possible. The agricultural sector is not exempt from this issue and has a significant share in the emission of greenhouse gases and global warming, the share of this sector is estimated at 10–12 % and the share of paddy fields at 0.11 % [38]. Considering the damage that fossil fuels cause to the environment, replacing these fuels with alternative energy sources including biofuels (fuels produced from renewable organic material) such as biodiesel may somehow reduce the carbon footprint [26]. However, it should be investigated whether it is possible to produce biofuels at a competitive cost with fossil fuels [27]. Basically, the emission of these gases occurs during agricultural activities directly (such as planting and harvesting operations), or indirectly during the production and transportation of farm inputs (such as herbicides, pesticides and fertilizers) [43]. In this regard, some researchers reported that a significant part of greenhouse gas emissions caused by the agricultural sector can be reduced by improving agricultural methods [6,18]. Therefore, optimizing the consumption of inputs leads to increased productivity and reduced greenhouse gas emissions [22]. The biggest factor in the emission of greenhouse gases into the atmosphere is the energy sector, and in recent years, along with the evaluation of energy in agricultural systems, the issue of greenhouse gas emissions has been of great interest [16]. In this

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way, the development of agricultural systems with less energy input and more productivity can help to reduce CO_2 greenhouse gas emissions, especially in the agricultural sector [11]. Most advanced and even developing countries have tried to optimize their agricultural systems in terms of energy consumption by examining the input energy at the surface for the production of various agricultural products and by calculating the energy efficiency index. Basavalingaiah et al. [4] investigated the determination of the pattern of energy use, greenhouse gas (GHG) emissions and efficiency of rice farms in puddled transplanted (PTR) and direct-seeded rice (DSR) production systems in India. According to the results, the mean input and output energy estimated for PTR were 35605 and 155776 MJ/ha and for DSR were 7823 and 53227 MJ/ha, respectively. Also, the results showed that the highest energy use efficiency (7.3), energy productivity (0.3 kg/MJ) and energy profitability (6.3) were related to DSR due to the lower energy input. The study by Bockari-Gevao et al. [6] in the field of energy consumption flow in Malaysia paddy fields showed that the highest average operational energy consumption in different stages of production was for tillage (1747.33 MJ/ha, 49 %) of the total operational energy consumption (3595.87 MJ/ha), followed by harvesting (33 %) and planting (16 %). Also, the average total input energy in the off and main cropping seasons was 12225.97 MJ/ha. According to energy sources, fertilizer and fuel recorded the highest energy consumption, which were 7721.03 and 2717.82 MJ/ha, respectively. Kosemani and Bamgboye [19] stated that the average energy input for rice production in small, medium and large farms in Nigeria were 14813, 14543 and 14067 MJ/ha, respectively, and the average energy output were 97546, 102864 and 107293 MJ/ha, respectively, and the average yield were 6695, 7060 and 7364 kg/ha, respectively. Energy input in fertilizer application was the highest with 74, 75 and 77 % of the total energy input in small, medium and large farms, respectively. The values of energy ratio in small, medium and large farms were 6.58, 7.07 and 7.62, energy productivity were 0.45, 0.48 and 0.52 kg/MJ, specific energy were 2.22, 2.08 and 1.92 MJ/kg, and net energy were 82733, 88321 and 93226 MJ/ha, respectively. Based on the results of Mardani Najafabadi et al. [25] on rice producers in Golestan province, Iran, all types of energy inputs and outputs were calculated as 34423.28 and 120088.4 MJ/ha, respectively. Nitrogen fertilizer (30 %), diesel (21 %), and irrigation water (21 %) had the highest values of energy inputs. The results of Yang et al. [46] on fully mechanized (FM) and semi-mechanized (SM) of rice production in China showed that the average energy input of FM and SM were 23610 and 26470.01 MJ/ha, respectively, whereas the rice and straw yields of FM were not significantly different from SM. Compared with SM, FM reduced fertilizer and water energy inputs by 1252.62 and 2187.87 MJ/ha, respectively. The energy use efficiency, energy productivity, and energy profitability of FM were 11.10, 0.41 kg/MJ, and 10.10, respectively, which increased by 8.6, 8.6, and 9.5 % compared to SM.

In addition, to meet the future food demand due to the global decrease in the availability of fresh water resources for agricultural production, the use of innovative water-saving rice production technologies is necessary to save a significant amount of water. Also, the water-loving nature of rice cultivation and scarcity of labor triggers the search for alternative crop establishment methods that can increase water productivity. The study of Bansal et al. [3] in Haryana state in India showed that rice grain yield by drip irrigation (6950 kg ha^{-1}) method was significantly increased compared to flood (6225 kg ha^{-1}) and sprinkler (4800 kg ha^{-1}) irrigation methods. Also, the drip and sprinkler irrigation methods of rice cultivation saved 30.7 and 28.7 % of water consumption, respectively, compared to the flood irrigation method. The results of Sharda et al. (2016) on drip irrigation in dry-seeded rice (DSR) and flood irrigation in South Asia showed that the grain yield in drip irrigation in DSR (7.34–8.01 t/ha) was higher than flood irrigation (6.63–7.60 t/ha), with water savings of more than 40 %. Also, water use efficiency with drip irrigation was higher (0.81–0.88 kg m^{-3}) than flood irrigation (0.42–0.52 kg m^{-3}). The study of Singh et al. [36] on two methods of direct-seeded rice cultivation and transplanted cultivation in two puddled and un-puddled beds in India indicated that puddling increased grain yield of rice by 0.7–1 ton.ha^{-1} and straw yield of rice 0.7–1.7 ton.ha^{-1}. According to the results, puddling reduced water requirement of rice by 75 mm ha^{-1}. Also, grain and straw yields in transplanted cultivation were significantly higher than direct-seeded rice cultivation in both puddled and unpuddled beds. Considering that the method of direct sowing of seeds and seedlings in dry bed has recently been proposed among farmers in Golestan province, knowing its environmental effects in Golestan province and similar climates (Mediterranean climate) is one of the most important questions of this planting method. In this context, it is hypothesized that variation in the method of cultivation and irrigation will lead to changes in GWP and energy efficiency. Thus, the objectives of this study were (i) to investigate the effect of different cultivation and irrigation methods on yield and water consumption and (ii) to study the change of energy consumption and global warming potential in these methods of rice cultivation and irrigation.

2. Materials and methods

2.1. Experimental location and design details

Field experiments were conducted at the Agricultural Research Station of Golestan province located in Gorgan city (36°45′ N, 54° 25′ E, 5.5 m above sea level), Golestan province, Iran, during 2019 and 2020. The current research was designed as split plot based on randomized complete block design with 12 treatments and three replications in 36 plots each measuring 6 m wide and 8 m length (48 m^2) in area of approximately 3600 m^2. The experiment was tested with modified rice variety of Fajr (high-yielding), which the growing period of the variety from sowing to the maturity stage is 130 days. Treatments including four systems of sprinkler irrigation (I_1) using four sprinklers for each plot (with a flow rate of 0.15 l/s per sprinkler), permanent flood irrigation (I_2), periodic flood irrigation (I_3) and drip (tape) irrigation (I_4) with 50 cm strip spacing (with 20 cm hole spacing and 1 l/h flow rate) were considered as main factors and three cultivation methods of direct-seeded rice (DSR) (C_1), un-puddled transplanted (C_2) and puddled transplanted (PTR) (C_3) were considered as sub-factors.

Before planting, the stages of land preparation including ploughing, leveling and plotting were done. For puddled transplanted treatments, the field was irrigated to a depth of 5 cm, then before leveling, It was puddled thoroughly using a four-wheel tractor. For un-puddled transplanted treatments, The puddling operation was not done and the seedling was directly placed in the soil after one irrigation stage. For two years of cultivation, 25-day-old seedlings were selected for transplantation. Planting was done by a field worker with planting distance of 25*25 cm and 4 to 5 seedlings per hill. Also, in direct-seeded rice treatments, the seeds were planted by tractor and along straight parallel furrows in lines 25 cm apart, at the same time as with transplanting. The plant population for transplanted and direct-seeded treatments was 80 plants/m^2. Also, the planting date for direct-seeded treatments were June 7, 2019 and June 6, 2020, for puddled transplanted treatments were June 10, 2019 and June 10, 2020 and for un-puddled transplanted treatments were June 12, 2019 and June 10, 2020.

The water needed for the experimental treatments was provided from the water well in the research station. In order to apply the irrigation treatments, the evaporation rate from the class-A evaporation pan was measured daily at 9:00 a.m. during the rice growth period. Then the evaporation from the pan was related to the reference evapotranspiration by using the experimental coefficient of the class-A evaporation pan in the form of the following relationship [2]:

$$ET_0 = K_p * E_{pan} \tag{1}$$

Where: ET_o = Reference evapotranspiration (mm/day), K_p = Coefficient of the pan (considered equal to 0.75) and E_{pan} = Pan evaporation (mm/

day). Then, rice plant evapotranspiration (ET_c) was calculated under standard conditions by multiplying the rice plant coefficient ($K_{c\ ini}$: 1.05, $K_{c\ mid}$: 1.2 and $K_{c\ end}$: 0.9) in reference evapotranspiration (ETo). Finally, the amount of irrigation water in each treatment was calculated based on the area of the plot and the percentage of evapotranspiration of the rice plant.

$$ET_c = ET_0 * K_c \tag{2}$$

For permanent flood irrigation, the water was continuously at a depth of approximately 5 cm on the soil surface, and in periodic flood irrigation, the plots were flooded with a depth of approximately 5 cm, and re-watering was done after the soil dried and capillary cracks were seen on the soil [6]. Information about the amount of water consumed by plots was obtained from the water meter (Multi-jet class B) that was installed for each treatment. The physical and chemical properties of the studied field soil are given in Table 1.

After the soil test, fertilizers were applied based on the nutrient status of the farm soil and technical recommendations (LCC[1]) according to rice needs. Basic fertilizers included urea (50 kg/ha), phosphorus (100 kg/ha) and potassium (100 kg/ha), which were applied equally to all treatments. After the physiological maturity of 50 % of the rice plants (hard and yellow colored grains and golden yellowing of leaves), irrigation was stopped. After the humidity of the plants reached the normal level, harvesting and threshing operations were done by the field worker after removing the margin. The number and duration of each operation, the amount of seed used in each plot, fertilizer and pesticide rates, and the amount of human labor involved in each operation (land preparation, planting, crop maintenance (including fertilization and weed control), threshing, and harvesting) were recorded through field measurements. The data were measured daily, and the calculated values are given in Tables 3 and 4.

2.2. Energy analysis

The energy equivalent coefficients for each of the inputs and outputs are shown in Table 2. To calculate the energy values of different treatments, all the measured data of the field, which were collected separately from the treatments, were entered into Excel software. Then, by multiplying the energy equivalents in the input and output values, the amount of energy available for each of the inputs and outputs was determined. Also, variance analysis of data was performed using SAS 9.2 statistical software and the comparison of means was performed using the least significant difference (LSD) test at the 5 % probability level.

In all stages of agricultural operations (including land preparation, planting and crop maintenance), the number of working hours in the field was recorded in terms of h/ha, then the total input energy of labor was obtained based on Equation (3) [13]:

$$E_{1a} = EI_{1a} * T * N \tag{3}$$

Where: E_{1a} = Energy of labor per hectare (MJ/ha), EI_{1a} = Equivalent energy of labor (MJ/h), T = Working hours of the labor per day (h/day), N = Number of labor, which is equivalent to the number of daily workers in the number of working days (day/ha).

The energy used in irrigation included both direct energy (electricity energy for transferring and distributing water) and indirect energy (energy related to the construction of pumps, pipes, pools and other devices), which was calculated by Equation (4):

$$E_w = W_C * \alpha \tag{4}$$

Where: E_w = Energy of irrigation (MJ/ha), W_c = Amount of water consumption (m^3/ha) and α = Coefficient of indirect irrigation energy, which was equal to 0.63 in fields with flood irrigation system [45] and in fields with pressurized irrigation system it was equal to 1.02 [1].

Other inputs such as fertilizers, pesticides, fuel and seed, used in the farms were transformed to energy value (MJ/ha) by multiplying the amount of the inputs applied per hectare by the energy equivalent of each input described in Table 2.

2.3. Energy indices

Based on energy inputs and outputs, energy indices such as energy use efficiency (energy ratio) (EUE), energy productivity (EP), specific energy (SE) and net energy (NE) were used to evaluate the performance of energy consumption in rice production, which were calculated by Equations (5)–(8) [39].

$$\mathrm{EUE} = \frac{\mathrm{E}_{out}\,(\mathrm{MJ/ha})}{\mathrm{E}_{in}\,(\mathrm{MJ/ha})} \tag{5}$$

$$\mathrm{EP} = \frac{\text{Rice yield (kg/ha)}}{\mathrm{E}_{in}\,(\mathrm{MJ/ha})} \tag{6}$$

$$\mathrm{SE} = \frac{\mathrm{E}_{in}\,(\mathrm{MJ/ha})}{\text{Rice yield (kg/ha)}} \tag{7}$$

$$\mathrm{NE} = \mathrm{E}_{out}\left(\frac{\mathrm{MJ}}{\mathrm{ha}}\right) - \mathrm{E}_{in}\left(\frac{\mathrm{MJ}}{\mathrm{ha}}\right) \tag{8}$$

2.4. Global warming potential

To calculate the global warming potential of rice production, input energy data from different treatments were used. Then, using the conversion coefficients of CO_2, N_2O and CH_4 gases equivalent to 1, 310 and 21 kg CO_2, respectively, the global warming potential due to greenhouse gas emissions was determined based on the equivalent of carbon dioxide (CO_2) [28].

GWP was calculated in four scales [39]:

GWP per unit area (per hectare) = GWP in terms of CO_2-equivalent per hectare.

GWP per unit weight = GWP per unit area (per hectare)/Total outputs (ton/ha)

GWP per unit input energy = GWP per unit area (per hectare)/Input energy (GJ/ha)

GWP per unit output energy = GWP per unit area (per hectare)/Output energy (GJ/ha)

3. Results and discussion

3.1. Inputs and outputs data in rice production

The machinery used (h/ha) in different stages of rice production are shown in Fig. 1. As can be observed from Fig. 1, the use of agricultural machinery was affected by the stages of land preparation. So, in the puddled transplanted cultivation, the use of machinery in the bed preparation stage increased due to puddling.

Also, the human labor used (h/ha) in different stages of rice production are shown in Fig. 2. As can be observed from Fig. 2, the human labor used in different stages of rice production was affected by the type of cultivation. So, the use of human labor for different irrigation systems in transplanted cultivation was higher than that of direct-seeded cultivation. The use of labor in the stage of bed preparation (ploughing, bordering, puddling, leveling, manuring) for the puddled transplanted treatment was more than that of un-puddled transplanted and direct-seeded treatments due to the need for puddling, which for the puddled transplanted, un-puddled transplanted and direct-seeded treatments were 19, 1.3 and 2 % for the sprinkler irrigation, 19.6, 9.9 and 14.1 % for the permanent flood irrigation, 19.4, 9.7 and 14.2 % for the periodic flood irrigation and 19.6, 1.4 and 2.1 % for the drip irrigation,

[1] Leaf Color Chart for N application.

Table 1
Physical and chemical properties of soil in the field (depth 30 cm).

pH	Electrical conductivity Ec*10^3	Available N %	Organic carbon O.C %	T.N.V lime %	P Available p.p.m	K Available p.p.m	Texture	Sand %	Silt %	Clay %
7.7	0.8	0.1	1.33	26.5	7.8	360	Si–C-L	20	52	28

Table 2
Energy equivalents of inputs and outputs in rice production.

Particulars (Unit)	Energy Equivalent (MJ $unit^{-1}$)	References
Inputs		
Labor (h)		
Male	1.96	[37]
Female	1.54	[37]
Machinery[a](kg)	142.7	[17]
Fuel (l)	37	[17]
Seed (kg)	14.7	[29]
Pesticides (kg)		
Herbicides	287	[33]
Fungicides	196	[12]
Insecticides	237	[33]
Chemical fertilizers (kg)		
Nitrogen (N)	60.6	[15]
Phosphorus (P_2O_5)	11.1	[15]
Potassium (K_2O)	6.7	[15]
Micronutrient (NPK)	120	[8]
Organic fertilizer (kg)	0.3	[29]
Electricity (kWh)	12.1	[29]
Irrigation water (m^3)		
Pressurized irrigation	1.02	[1]
Flood irrigation	0.63	[15]
Outputs		
Rice straw (kg)	12.5	[29]
Rice grain (kg)	14.7	[29]

[a] This energy includes construction, repairs, maintenance and transportation.

respectively.

In the planting stage, the use of labor for transplanted cultivation was more than the direct-seeded cultivation due to the need for labor for transplanting. Also, in the holding stage (fertilizing, spraying, weeding, irrigation), the average use of labor for the direct-seeded and un-puddled transplanted cultivation was higher than that of puddled transplanted cultivation due to the need for weeding. Sharma et al. [35] reported that the direct-seeded rice cultivation eliminates the need for puddling, thus reducing water and labor demands. In most paddy fields under direct-seeded rice cultivation in Asia and Africa, weeding is done 2–3 times during the season and for this purpose, more than 100 person-days per hectare of labor are used [34].

The amount and type of inputs and outputs in rice production per hectare are given in Tables 3 and 4 According to the results, the amount of fertilizers used during the rice growth period, for the transplanted and direct-seeded rice treatments for the first year were 260.4 and 270.8 kg/ha nitrogen, 156.25 and 208.3 kg/ha phosphorus and 208.3 and 197.9 kg/ha potassium, respectively, and for the second year were 229.17 and 260.4 kg/ha nitrogen, 156.25 and 156.25 kg/ha phosphorus and 166.67 and 166.67 kg/ha potassium, respectively. The results showed that the need for nitrogen fertilizer for direct-seeded rice cultivation, due to the longer growth period of rice seeds in the field was more than transplanted cultivation (which was also reported by Mahajan et al. [21]).

As can be observed (Tables 3 and 4), the lowest amount of water consumption was related to the drip irrigation treatment with direct-seeded rice cultivation (6388.9 and 6296.89 m^3/ha for the first and second years, respectively). Also, the highest amount of water consumption was related to the permanent flood irrigation treatment with direct-seeded rice cultivation (12784.7 and 12189.3 m^3/ha for the first and second years, respectively), which occurred due to the longer growth period of rice seeds in the field and the need for more frequent of irrigation as well as the high volume of water consumed in the flood irrigation system, which was also reported by Mahajan et al. [20]. Chauhan and Opena [10] reported that puddling in the flood irrigation system of transplanted rice, consumes up to 30 % of the total water required of rice. Also, the results showed (Tables 3 and 4) that the transplanted cultivation had a higher yield than direct-seeded cultivation under all irrigation systems. According to the results, the highest amount of yield was obtained for the permanent flood irrigation system with puddled transplanted cultivation, which were 8051.4 and 8360 kg/ha for the first and second years, respectively. Also, the lowest amount of yield was for sprinkler irrigation system with direct-seeded cultivation, which were 2819.3 and 3302.9 kg/ha for the first and second years, respectively. The results of this study showed that un-puddled transplanted cultivation had a lower yield reduction compared to puddled transplanted cultivation, while direct-seeded cultivation had a large yield decrease. However, it should be investigated whether the use

Table 3
The amount of inputs and outputs for different treatments of rice production per hectare (2019).

Items	I_1				I_2			I_3			I_4	
	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3
Inputs (Unit)												
Labor (h)	525.3	790.1	813.6	580.7	830.2	786.7	583.6	833.9	792.83	511.7	763.34	788.92
Machinery (h)	124.4	123.24	265.02	198.2	193.2	270.01	195.7	192.1	268.76	124.24	123.1	268
Fuel (l)	123.1	112.53	138.4	136.9	126.3	138.48	136.8	126.2	138.46	123.1	112.53	138.4
Transportation (km)	5	4.5	5.5	5.5	5	5.5	5.5	5	5.5	5	4.5	5.5
Seed (kg)	60	100	100	60	100	100	60	100	100	60	100	100
Pesticides (kg)												
Herbicides	3.58	1.14	1.14	3.58	1.14	1.14	3.58	1.14	1.14	3.58	1.14	1.14
Fungicides	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
Insecticides	4.49	66.99	66.99	4.49	66.99	66.99	4.49	66.99	66.99	4.49	66.99	66.99
Chemical fertilizers (kg)												
Nitrogen (N)	270.8	260.4	260.4	270.8	260.4	260.4	270.8	260.4	260.4	270.8	260.4	260.4
Phosphorus (P_2O_5)	208.3	156.25	156.25	208.3	156.25	156.25	208.3	156.25	156.25	208.3	156.25	156.25
Potassium (K_2O)	197.9	208.3	208.3	197.9	208.3	208.3	197.9	208.3	208.3	197.9	208.3	208.3
Micronutrient (NPK)	0.25	0.25	0.25	–	–	–	–	–	–	–	–	–
Irrigation water (m^3)	7229.17	7458.3	9854.2	12784.7	11937.5	12472.2	10958.3	10305.6	11562.5	6388.9	6715.28	9173.6
Outputs (Unit)												
Rice straw (kg)	7937.05	8916.6	8134.8	10376.9	7698.9	10364.6	9577.6	8294.6	9118.7	9540.9	10053.9	11269.1
Rice grain (kg)	2819.3	4530.96	6888.1	4684.3	7021.8	8051.4	4229.47	6629.27	7487.27	4187.5	6323.8	7422.3

Table 4
The amount of inputs and outputs for different treatments of rice production per hectare (2020).

	I_1			I_2			I_3			I_4		
	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3
Inputs (Unit)												
Labor (h)	495.9	734.4	789.2	545.98	789.92	772.88	544.63	803.9	781.84	473.75	724.04	769.91
Machinery (h)	118.67	116.8	254.77	189.5	186.3	255.5	189.53	186.3	255.53	118.7	116.8	254.74
Fuel (l)	109.9	99.1	124.8	123.6	112.8	124.82	123.6	112.83	124.8	109.92	99.1	124.8
Transportation (km)	4	3.5	4.5	4.5	4	4.5	4.5	4	4.5	4	3.5	4.5
Seed (kg)	60	100	100	60	100	100	60	100	100	60	100	100
Pesticides (kg)												
Herbicides	4.04	0.14	0.14	4.04	0.14	0.14	4.04	0.14	0.14	4.04	0.14	0.14
Fungicides	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
Insecticides	1.46	1.46	1.46	1.46	1.46	1.46	1.46	1.46	1.46	1.46	1.46	1.46
Chemical fertilizers (kg)												
Nitrogen (N)	260.4	229.17	229.17	260.4	229.17	229.17	260.4	229.17	229.17	260.4	229.17	229.17
Phosphorus (P_2O_5)	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25
Potassium (K_2O)	166.67	166.67	166.67	166.67	166.67	166.67	166.67	166.67	166.67	166.67	166.67	166.67
Micronutrient (NPK)	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04
Organic fertilizer (kg)	7.34	7.34	7.34	7.34	7.34	7.34	7.34	7.34	7.34	7.34	7.34	7.34
Irrigation water (m^3)	6908.2	7158.3	8631.5	12189.3	11268.2	11461.3	10784.7	10193.7	10475.2	6296.9	6427.8	8037.1
Outputs (Unit)												
Rice straw (kg)	8702.8	9249.1	8244.7	9555	8548	10396.6	10477.7	9521.8	8707.5	9034.5	8745	9753.5
Rice grain (kg)	3302.9	5063.35	7167.9	5320.5	7956.6	8360	5324.8	7803.4	7407.5	4407.6	6205.9	7264.6

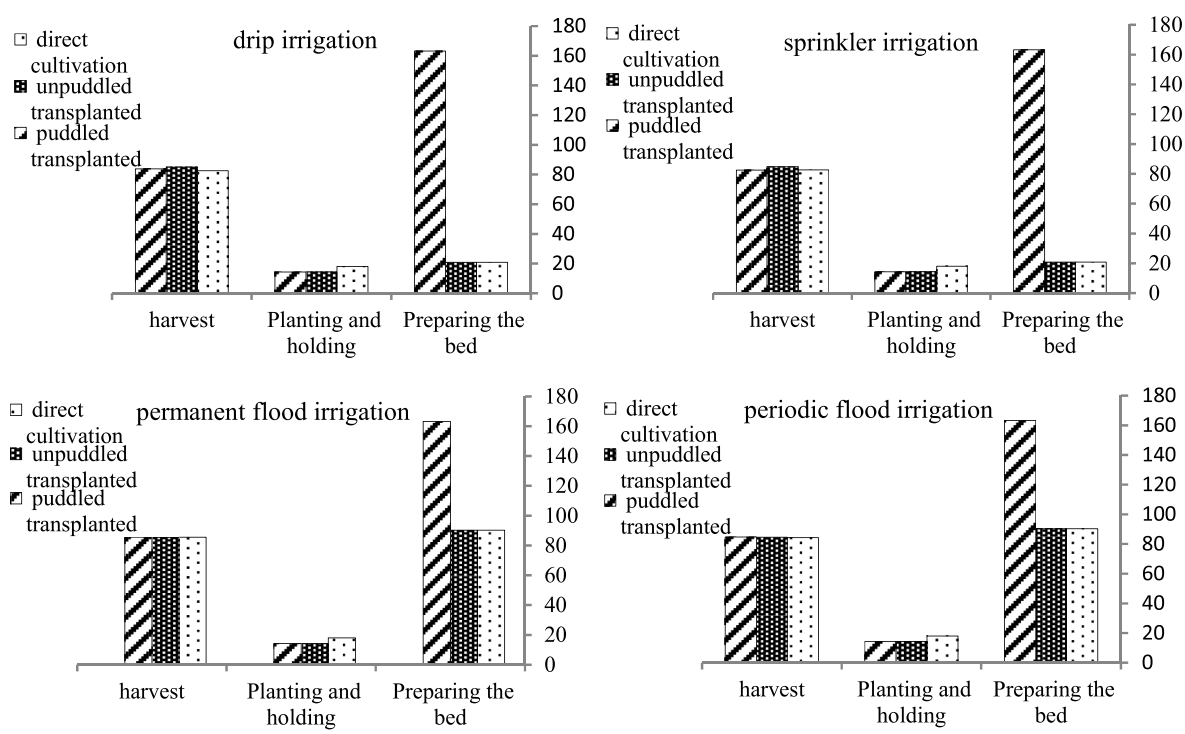

Fig. 1. The machinery used (h/ha) in different stages of rice production.

of a drip irrigation system is cost-effective or not. The results of the study of Bouman and Tuong [7] showed that most water-saving technologies, including aerobic rice cultivation, lead to a decrease in rice yield. Xu et al. [44], also reported that the yield of direct-seeded rice cultivation was 12 % lower than that of transplanted puddled rice cultivation. As can be seen (Tables 3 and 4), under different irrigation and cultivation scenarios, drip irrigation with UPTR cultivation, can be a technology to save water (46 and 44 % saving of water consumption for the first and second year, respectively) with yield preservation (21 and 26 % yield reduction for the first and second year, respectively) compared to permanent flood irrigation with PTR cultivation for areas with water crisis.

3.2. Input-output energy in rice production

The amount and percentage of energy inputs and outputs (MJ/ha) of rice production are given in Tables 5–7. The energy input of different irrigation systems showed that the energy input for puddled transplanted treatment was higher than un-puddled transplanted and direct-seeded cultivation treatments, which was related to the higher energy consumption of machinery in the land preparation stage.

As can be seen (Tables 5–7), the amount of energy consumption by nitrogen fertilizer was higher than phosphorus and potassium fertilizer, which was also reported by Eskandari Cherati et al. [14] and Mansoori et al. [24] in Iran.

The results of Tables 5–7 showed that the highest share of input energy consumption for different irrigation and cultivation treatments

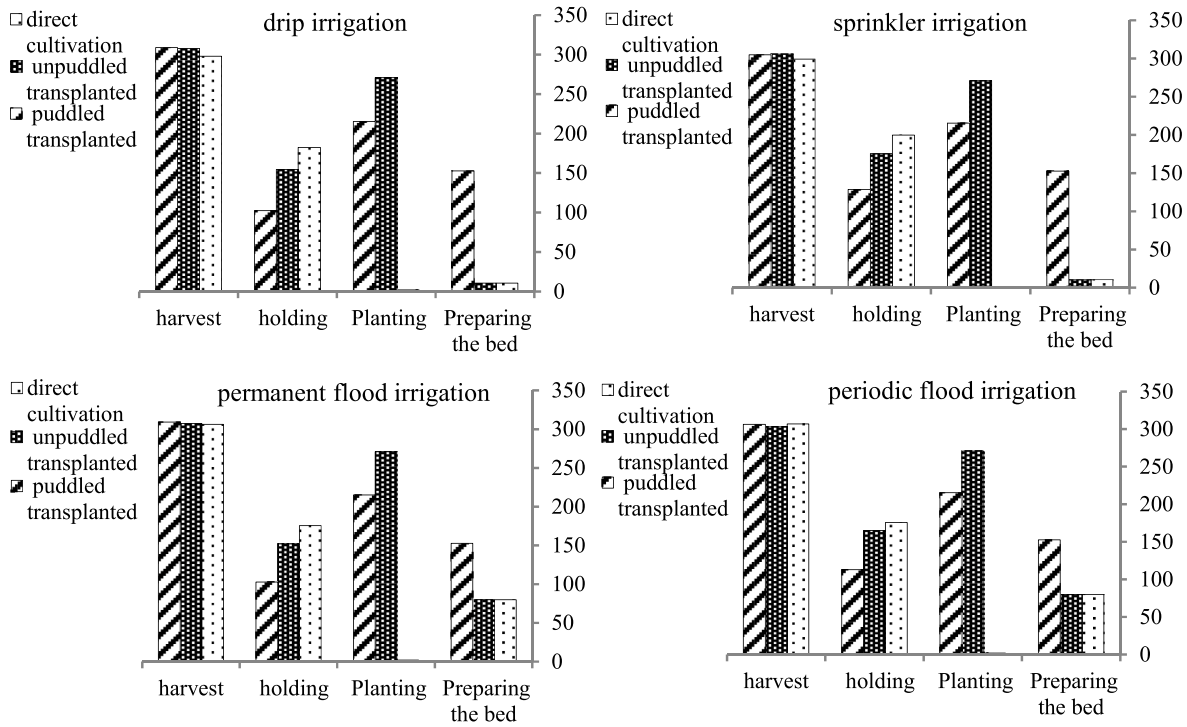

Fig. 2. The human labor used (h/ha) in different stages of rice production.

was related to irrigation water (between 21 and 29 %), nitrogen fertilizer (between 20 and 28 %) and fuel (between 14 and 17 %), similar to that reported in India [9]. Also, the input energy consumption of machinery, electricity, labor and seeds were in the next categories.

Bockari-Gevao et al. [6], also stated that the major share of input energy was related to chemical fertilizers. Also, Mani and Patel [23] reported that the highest energy consumption in different farms in India was related to electrical and fertilizer. Pishgar et al. [32] reported that the amount of energy consumption in rice production in Guilan province of Iran was 39333 MJ/ha, which 46 % was related to diesel fuel and 36 % was related to chemical fertilizers. Wakil et al. [42] stated that the total energy consumption in rice cultivation was 36397.85 MJ/ha, that the largest share of energy consumption was related to irrigation water (48.51 %) and fuel (23.58 %). The results of Yang et al. [46] on rice production in China showed that the highest share of energy consumption was related to fertilizer, fuel, and water, which included 92.02 % of the total energy input.

3.2.1. *Variance analysis of energy indices of rice production*

The results of variance analysis and mean comparison of energy input and output and the indices of energy use efficiency (EUE), energy productivity (EP), specific energy (SE) and net energy (NE) for different irrigation and cultivation treatments of rice production are given in Tables 8 and 9. Based on the results of analysis of variance for all investigated traits (Table 8), irrigation and cultivation methods were significant at the 1 % level.

The results of mean comparison of irrigation method (Table 9) showed that the highest amount of energy input (30524 MJ/ha) and output (220042 MJ/ha) were obtained for the permanent flood irrigation system and the lowest amount of energy input (29076 MJ/ha) and output (179578 MJ/ha) were obtained for the sprinkler irrigation system. Although the sprinkler irrigation system reduced the amount of energy input, it also significantly reduced the yield, which reduced the amounts of EP (0.168 kg/MJ), EUE (6.22) and NE (150502 MJ/ha). Also, the highest amount of EUE was obtained for the periodic flood irrigation system (7.25), but there was not a significant difference with treatments of permanent flood irrigation and drip irrigation, and the highest amount of EP was obtained for the permanent flood irrigation system (0.223 kg/MJ).

The results of mean comparison of the cultivation method (Table 9) showed that the highest and lowest amount of energy input were obtained for the treatments of puddled transplanted (32418 MJ/ha) and un-puddled transplanted (28175 MJ/ha), respectively. As can be seen, there were no significant difference in amounts of EP and EUE between un-puddled transplanted (7.27 and 0.227 kg/MJ) and puddled transplanted (7.11 and 0.233 kg/MJ) treatments, due to lower input energy in un-puddled transplanted treatment. Also, the highest and lowest amounts of energy output were obtained for the treatments of puddled transplanted (229074 MJ/ha) and direct-seeded (180487 MJ/ha), respectively. The lowest amounts of EUE (6.37), EP (0.151 kg/MJ) and NE (152172 MJ/ha) were obtained for the direct-seeded treatment. The highest amount of SE was obtained for the direct-seeded cultivation (6.91 MJ/kg), and the lowest amount of SE was obtained for the puddled transplanted cultivation (4.37 MJ/kg) but there was not a significant difference with un-puddled transplanted cultivation.

The study by Than et al. [41] on rice cultivation showed the average total energy input in transplanted and direct seeded treatments were 19501.20 and 19224.58 MJ ha^{-1}, and the average total energy output was 166781.7 and 160445.04 MJ ha^{-1}, respectively. Also, the amounts of EUE, SE and EP in transplanted cultivation were 8.55, 2.89 MJ kg^{-1} and 0.35 kg MJ^{-1}, respectively, and in direct-seeded rice cultivation were 8.35, 3.31 MJ kg^{-1} and 0.3 kg MJ^{-1}, respectively. Wakil et al. [42] stated that the amounts of EUE, SE, EP, water productivity and NE in puddled transplanted rice cultivation were 2.47, 31.38 MJ/kg, 0.147 kg/MJ, 0.103 kg/m^3 and 53598.71 MJ/ha, respectively. The study by Yuan and Peng [47] on rice cultivation showed that the values of input energy, output energy, EUE, EP, SE and NE in farmers' practice (FP) were 34544.4 MJ/ha, 266744.9 MJ/ha, 7.77, 0.27 kg/MJ, 3.76 MJ/kg and 232200.5 MJ/ha, respectively, and in simplified and reduced-input practice (SRIP), were 25441.5 MJ/ha, 234393 MJ/ha, 9.25, 0.33 kg/MJ, 3.01 MJ/kg and 208951.5 MJ/ha, respectively.

Table 5
The amount and percentage of energy inputs and outputs (MJ/ha) in direct-seeded rice cultivation method.

Items	I_1				I_2				I_3				I_4			
	2019	%	2020	%	2019	%	2020	%	2019	%	2020	%	2019	%	2020	%
Inputsrowhead																
Labor	1029.6	3.7	974.8	3.82	1138.1	3.62	1070.1	3.69	1143.9	3.82	1067.5	3.85	1002.9	3.51	928.5	3.51
Machinery	1931.3	6.93	1678.9	6.6	3510.7	11.16	3251.3	11.21	3504.7	11.72	3251.3	11.72	1931	6.75	1678.9	6.34
Fuel	4677.8	16.78	4176.8	16.4	5202.2	16.54	4697	16.2	5199.7	17.38	4697.4	16.93	4678.4	16.35	4176.8	15.79
Transportation	224.7	0.81	176.8	0.7	246.4	0.78	198.5	0.68	246.4	0.82	198.5	0.72	224.7	0.78	176.8	0.67
Seed	882	3.16	882	3.46	882	2.8	882	3.04	882	2.95	882	3.18	882	3.08	882	3.33
Pesticidesrowhead																
Herbicides	467.7	1.68	384.91	1.51	467.7	1.49	384.91	1.33	467.7	1.56	384.91	1.39	467.7	1.63	384.91	1.45
Fungicides	38.7	0.14	38.7	0.15	38.7	0.12	38.7	0.13	38.7	0.13	38.7	0.14	38.7	0.14	38.7	0.15
Insecticides	469.91	1.69	2.07	0.01	469.91	1.49	2.07	0.01	469.91	1.57	2.07	0.01	469.91	1.64	2.07	0.01
Chemical fertilizersrowhead																
Nitrogen (N)	7549.75	27.09	7259.4	28.49	7549.75	24	7259.4	25.03	7549.75	25.24	7259.4	26.17	7549.75	26.39	7259.4	27.44
Phosphorus (P_2O_5)	1063.75	3.82	797.8	3.13	1063.75	3.38	797.8	2.75	1063.75	3.56	797.8	2.88	1063.75	3.72	797.8	3.02
Potassium (K_2O)	636.5	2.28	536	2.1	636.5	2.02	536	1.85	636.5	2.13	536	1.93	636.5	2.22	536	2.03
Micronutrient (NPK)	13.1	0.05	57.5	0.23	–	–	57.5	0.2	–	–	57.5	0.21	–	–	57.5	0.22
Organic fertilizer	–	–	1.01	0.0006	–	–	1.01	0.0005	–	–	1.01	0.0005	–	–	1.01	0.0006
Electricity	1512.5	5.43	1468	5.76	2193.1	6.97	2142.7	7.39	1802.4	6.03	1769.6	6.38	3146	11	3115.7	11.78
Irrigation water	7373.7	26.45	7046.3	27.68	8054.4	25.61	7679.2	26.48	6903.7	23.08	6794.4	24.49	6516.7	22.78	6422.8	24.27
Outputsrowhead																
Rice straw	99212.9	70.54	108785.4	69.14	129711.2	65.32	119437.5	60.43	119720.8	65.82	130971.7	62.59	119261.3	65.96	112931.7	63.54
Rice grain	41443.7	29.46	48552.97	30.86	68859.7	34.68	78211	39.57	62173.2	34.18	78274.1	37.41	61556.7	34.04	64791.9	36.46

Table 6
The amount and percentage of energy inputs and outputs (MJ/ha) in un-puddled transplanted cultivation method of rice.

Items	I_1				I_2				I_3				I_4			
	2019	%	2020	%	2019	%	2020	%	2019	%	2020	%	2019	%	2020	%
Inputsrowhead																
Labor	1548.6	5.29	1439.5	5.77	1627.2	5.12	1548.2	5.65	1634.5	5.38	1575.6	5.97	1496.1	5.03	1419.1	5.57
Machinery	1777.9	6.07	1493.7	6	3348.1	10.53	3062.7	11.17	3345.5	11.01	3062.7	11.61	1777.5	5.98	1493.7	5.86
Fuel	4276.3	14.6	3767.1	15.11	4799.4	15.09	4287.7	15.64	4796.9	15.78	4287.7	16.25	4276.3	14.38	3767.1	14.78
Transportation	200.4	0.68	152.5	0.61	222.2	0.7	174.3	0.64	222.2	0.73	174.3	0.66	200.4	0.67	152.5	0.6
Seed	1470	5.02	1470	5.9	1470	4.62	1470	5.36	1470	4.84	1470	5.57	1470	4.94	1470	5.77
Pesticidesrowhead																
Herbicides	130.6	0.45	15.55	0.06	130.6	0.41	15.55	0.06	130.6	0.43	15.55	0.06	130.6	0.44	15.55	0.06
Fungicides	38.7	0.13	38.7	0.16	38.7	0.12	38.7	0.14	38.7	0.13	38.7	0.15	38.7	0.13	38.7	0.15
Insecticides	1951.2	6.66	2.07	0.01	1951.2	6.13	2.07	0.01	1951.2	6.42	2.07	0.01	1951.2	6.56	2.07	0.01
Chemical fertilizersrowhead																
Nitrogen (N)	7259.4	24.79	6388.25	25.63	7259.4	22.82	6388.25	23.3	7259.4	23.88	6388.25	24.21	7259.4	24.42	6388.25	25.07
Phosphorus (P_2O_5)	797.8	2.72	797.8	3.2	797.8	2.51	797.8	2.91	797.8	2.62	797.8	3.02	797.8	2.68	797.8	3.13
Potassium (K_2O)	670	2.29	536	2.15	670	2.11	536	1.95	670	2.2	536	2.03	670	2.25	536	2.1
Micronutrient (NPK)	13.8	0.05	57.5	0.23	–	–	57.5	0.21	–	–	57.5	0.22	–	–	57.5	0.23
Organic fertilizer	–	–	1.01	0.0005	–	–	1.01	0.0005	–	–	1.01	0.0004	–	–	1.01	0.0005
Electricity	1545.6	5.28	1462.1	5.87	1971.3	6.2	1942.7	7.08	1589.8	5.23	1553.7	5.9	2813.2	9.46	2783	10.92
Irrigation water	7607.5	25.97	7301.5	29.3	7520.6	23.64	7099	25.9	6492.5	21.36	6422.1	24.34	6849.6	23.04	6556.3	25.73
Outputsrowhead																
Rice straw	111457.5	62.59	115614.2	60.84	96236.2	48.25	106849.6	47.74	103682.5	51.55	119022.1	50.92	125674.2	57.48	109312.9	54.51
Rice grain	66605.2	37.41	74431.2	39.16	103221	51.75	116961.5	52.26	97450.2	48.45	114710	49.08	92959.4	42.52	91226.2	45.49

Table 7
The amount and percentage of energy inputs and outputs (MJ/ha) in puddled transplanted cultivation method of rice.

Items	I_1			I_2			I_3			I_4		
	2019	%	2020	2019	%	2020	2019	%	2020	2019	%	2020	%
Inputsrowhead													
Labor	1594.6	4.41	1546.86	1542	4.52	1514.8	1553.9	4.68	1532.4	1546.3	4.25	1509	4.87
Machinery	5063.2	14.02	4629.5	5075.1	14.9	4631.2	5072.1	15.3	4631.3	5070.2	13.94	4629.4	14.93
Fuel	5259.2	14.56	4743	5262.4	15.44	4743	5261.7	15.9	4743	5259.8	14.46	4742.4	15.29
Transportation	243.9	0.67	196	243.9	0.71	196	243.9	0.73	196	243.9	0.67	196	0.63
Seed	1470	4.07	1470	1470	4.31	1470	1470	4.43	1470	1470	4.04	1470	4.74
Pesticidesrowhead													
Herbicides	130.6	0.36	15.55	130.6	0.38	15.55	130.6	0.39	15.55	130.6	0.36	15.55	0.05
Fungicides	38.7	0.11	38.7	38.7	0.11	38.7	38.7	0.12	38.7	38.7	0.11	38.7	0.12
Insecticides	1951.2	5.4	2.07	1951.2	5.72	2.07	1951.2	5.9	2.07	1951.2	5.37	2.07	0.01
Chemical fertilizersrowhead													
Nitrogen (N)	7259.4	20.1	6388.25	7259.4	21.3	6388.25	7259.4	21.9	6388.25	7259.4	19.96	6388.25	20.61
Phosphorus (P_2O_5)	797.8	2.21	797.8	797.8	2.34	797.8	797.8	2.41	797.8	797.8	2.19	797.8	2.57
Potassium (K_2O)	670	1.85	536	670	1.97	536	670	2.02	536	670	1.84	536	1.73
Micronutrient (NPK)	13.8	0.04	57.5	–	–	57.5	–	–	57.5	–	–	57.5	0.19
Organic fertilizer	–	–	1.01	–	–	1.01	–	–	1.01	–	–	1.01	0.0004
Electricity	1577.7	4.37	1549.6	1789.8	5.25	1764.6	1436.9	4.33	1441.9	2571.2	7.07	2420	7.81
Irrigation water	10051.2	27.83	8804.1	7857.5	23.05	7220.6	7284.4	21.95	6599.4	9357.1	25.73	8197.8	26.44
Outputsrowhead													
Rice straw	101685.8	50.11	103058.7	129558.3	52.26	129957.5	113983.7	50.88	108844.2	140863.8	56.35	121918.7	53.31
Rice grain	101255.1	49.89	105367.6	118355.1	47.74	122892	110062.8	49.12	108889.7	109107.3	43.65	106790	46.69

3.3. Global warming potential (GWP) caused by rice production

The values of global warming potential of different treatments per unit area (kg CO_2 equivalent per hectare) caused by different activities are given in Tables 10 and 11. According to the results, for different irrigation systems, puddled transplanted cultivation had a higher amount of GWP per unit area (per hectare) (which was reported in India [31]) compared to un-puddled transplanted and direct-seeded cultivation, due to the higher use of machinery for land preparation. The highest amount of GWP per unit area (per hectare) were 2217.52 and 1814.79 kg CO_2 equivalent per hectare for the first and second years, respectively, which was related to the puddled transplanted cultivation with drip irrigation system. Also, a similar report in India showed that there was a significant reduction in total greenhouse gas emissions in the direct-seeded rice (574.1 kg CO_2 eq./ha) compared to the puddled transplanted rice cultivation (3954.8 kg CO_2 eq./ha) [3]. The results of Pathak and Wassmann [30] in India showed that GWP for rice crop was in the range of 2766–4054 kg CO_2 equivalent per hectare. Soni et al. [40] reported that puddled transplanted rice cultivation had the highest amount of greenhouse gas emissions (1112 kg CO_2 eq./ha) compared to other crops.

Based on the results of Tables 10 and 11, the largest share of GWP in rice cultivation was related to the use of nitrogen fertilizer, fuel and machinery, similar to that reported in India [4,31].

3.3.1. Variance analysis of global warming potential (GWP) of rice production

The results of variance analysis and mean comparison of global warming potential of different treatments per unit area (kg CO_2 equivalent per hectare), weight (kg CO_2 equivalent per ton), input energy (kg CO_2 equivalent per GJ), and output energy (kg CO_2 equivalent per GJ) are given in Tables 12 and 13. Based on the results of variance analysis of GWP per unit area and weight (Table 12), irrigation, cultivation method, the interaction effect of cultivation method* irrigation and cultivation method* year were significant at the 1 % level, and the interaction effect of cultivation method* irrigation* year were not significant. The results of variance analysis of GWP per unit input energy showed that irrigation and cultivation methods were significant at the 1 % level. Also, the results of variance analysis of GWP per unit output energy showed that irrigation and cultivation method* year were significant at the 5 % level and cultivation method and the interaction effect of cultivation method* irrigation were significant at the 1 % level.

According to the results of mean comparison of irrigation method (Table 13), the lowest amount of GWP per unit area (1608 kg CO_2 equivalent per hectare) and input energy (55 kg CO_2 equivalent per GJ) were obtained for the sprinkler irrigation system, due to lower energy input. Also, the lowest amount of GWP per unit weight was obtained for the drip irrigation system (111.9 kg CO_2 equivalent per ton) but there was not a significant difference with treatments of periodic flood irrigation and permanent flood irrigation, and the lowest amount of GWP per unit output energy was obtained for the permanent flood irrigation system (8.3 kg CO_2 equivalent per GJ) but there was not a significant difference with treatments of periodic flood and drip irrigation.

The results of mean comparison of cultivation method (Table 13) showed that the lowest amount of GWP per unit area (1593 kg CO_2 equivalent per hectare), weight (104.8 kg CO_2 equivalent per ton), input energy (56.3 kg CO_2 equivalent per GJ) and output energy (7.8 kg CO_2 equivalent per GJ) were obtained for the un-puddled transplanted cultivation, which was due to lower energy input and a yield close to puddled transplanted cultivation.

4. Conclusions

This study evaluated the energy consumption and global warming potential for transplanted and direct-seeded rice cultivation methods under sprinkler, flood and drip irrigation systems. The results showed

Table 8
Two-year variance analysis of energy indices in rice production per hectare.

Source of variance	Degree of Freedom (df)	Energy input	Energy output	EUE	EP	SE	NE
Year	1	**267935043	679048741ns	**21.02	**0.0308	**22.62	1800074530ns
Replication in year	4	12783	1184387136	1.366	0.0002	0.294	1178790503
Irrigation	3	**7148489	**5588986168	**4.053	**0.0117	**12.22	**5289336170
Irrigation*Year	3	**132669	882003468ns	0.721ns	0.0005ns	0.704ns	672787345ns
Main error	12	16803	308677903	0.653	0.0005	0.382	509350805
Cultivation method	2	**139486471	**14170502721	**5.565	**0.0504	**48.07	**11948492325
Cultivation method*Irrigation	6	**12929196	596620106ns	0.976ns	0.0008ns	**1.826	506581318ns
Cultivation method*Year	2	**11976901	570557906ns	0.562ns	0.0005ns	*1.338	417922394ns
Cultivation method* Irrigation*Year	6	**79669	406481955ns	0.309ns	0.0001ns	0.177ns	306225997ns
error	32	15768	271126393	0.522	0.0004	0.358	471384748
Coefficient of Variation (%) (CV)	-	4.6	11.6	2.2	1.8	3.3	13.6

ns, * and **: non-significant and significant at 5% and 1% levels of probability, respectively.

Table 9
Mean comparison of irrigation and cultivation method on energy indices in rice production.

traits		Energy input	Energy output	EUE	EP	SE	NE
Year	2019	31565[a]	202008[a]	6.38[b]	0.183[b]	5.84[a]	170443[a]
	2020	27707[b]	208150[a]	7.46[a]	0.224[a]	4.72[b]	180443[a]
Cultivation method	Direct-seeded cultivation	28314[b]	180487[c]	6.37[b]	0.151[b]	6.91[a]	152172[c]
	Un-puddled transplanted	28175[c]	205677[b]	7.27[a]	0.227[a]	4.55[b]	177501[b]
	Puddled transplanted	32418[a]	229074[a]	7.11[a]	0.233[a]	4.37[b]	196655[a]
Irrigation	Sprinkler irrigation	29076[d]	179578[b]	6.22[b]	0.168[c]	6.48[a]	150502[b]
	Permanent flood	30524[a]	220042[a]	7.13[a]	0.223[a]	4.71[c]	189518[a]
	Periodic flood	29337[c]	211298[a]	7.25[a]	0.222[a]	4.78[bc]	181960[a]
	Drip irrigation	29607[b]	209399[a]	7.08[a]	0.201[b]	5.14[b]	179792[a]

Means with the similar letters in each column are not significantly different at 5 % of probability level (LSD Test).

Table 10
Values of GWP per unit area (kg CO_2 equivalent per hectare) caused by different activities for different treatments (2019).

Items	I_1			I_2			I_3			I_4		
	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3
Inputs												
Nitrogen (N)	549.73	528.58	528.58	549.73	528.58	528.58	549.73	528.58	528.58	549.73	528.58	528.58
Phosphorus (P_2O_5)	87.23	65.42	65.42	87.23	65.42	65.42	87.23	65.42	65.42	87.23	65.42	65.42
Potassium (K_2O)	52.19	54.94	54.94	52.19	54.94	54.94	52.19	54.94	54.94	52.19	54.94	54.94
Pesticides	100.37	217.99	217.99	100.37	217.99	217.99	100.37	217.99	217.99	100.37	217.99	217.99
Machinery	241.59	222.41	633.38	439.17	418.83	634.87	438.42	418.5	634.49	241.56	222.36	634.25
Transportation	17.53	15.63	19.02	19.22	17.33	19.02	19.22	17.33	19.02	17.53	15.63	19.02
Fuel:												
Agricultural operations	364.87	333.55	410.22	405.77	374.35	410.47	405.58	374.16	410.41	364.91	333.55	410.26
Irrigation[a]	168.85	172.55	176.14	244.83	220.07	199.81	201.22	177.48	160.41	351.22	314.06	287.05
GWP												
GWP per unit area	1582.36	1611.07	2105.69	1898.51	1897.52	2131.1	1853.95	1854.4	2091.28	1764.73	1752.54	2217.53

[a] Fuel for generating electricity or use for pumping water.

Table 11
Values of GWP per unit area (kg CO_2 equivalent per hectare) caused by different activities for different treatments (2020).

Items	I_1			I_2			I_3			I_4		
	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2	C_3
Inputs												
Nitrogen (N)	528.6	465.15	465.15	528.6	465.15	465.15	528.6	465.15	465.15	528.6	465.15	465.15
Phosphorus (P_2O_5)	65.42	65.42	65.42	65.42	65.42	65.42	65.42	65.42	65.42	65.42	65.42	65.42
Potassium (K_2O)	43.95	43.95	43.95	43.95	43.95	43.95	43.95	43.95	43.95	43.95	43.95	43.95
Pesticides	43.76	5.79	5.79	43.76	5.79	5.79	43.76	5.79	5.79	43.76	5.79	5.79
Machinery	210.02	186.85	579.12	406.72	383.13	579.34	406.72	383.13	579.35	210.02	186.85	579.11
Transportation	13.79	11.9	15.29	15.48	13.59	15.29	15.48	13.59	15.29	13.79	11.9	15.29
Fuel:												
Agricultural operations	325.79	293.83	369.95	366.37	334.44	369.95	366.4	334.44	369.95	325.79	293.83	369.91
Irrigation[a]	163.89	163.23	173	239.21	216.88	197	197.56	173.45	160.97	347.83	310.69	270.17
GWP												
GWP per unit area	1395.2	1236.12	1717.68	1709.5	1528.36	1741.89	1667.87	1484.93	1705.88	1579.15	1383.59	1814.79

[a] Fuel for generating electricity or use for pumping water.

Table 12
Two-year variance analysis for GWP of rice per hectare.

Source of variance	Degree of Freedom (df)	GWP per unit area	GWP per unit weight	GWP per unit input energy	GWP per unit output energy
Year	1	**1800575	**10592	**173	**57.2
Replication in year	4	22091	131	22	0.8
Irrigation	3	**143464	**284	**96	*2.1
Irrigation*Year	3	22^{ns}	**313	0.17^{ns}	1.4^{ns}
Main error	12	12290	37	12	0.35
Cultivation method	2	**781772	**2244	**86	**15.5
Cultivation method* Irrigation	6	**41626	**241	2.7^{ns}	**1.31
Cultivation method*Year	2	**75916	**185	12^{ns}	*1.3
Cultivation method* Irrigation*Year	6	39^{ns}	57^{ns}	0.37^{ns}	0.4^{ns}
error	32	11144	30	10.5	0.34
Coefficient of Variation					
CV (%)	-	6.1	4.8	6.1	6.8

^{ns}, * and **: non-significant and significant at 5% and 1% levels of probability, respectively.

Table 13
Mean comparison of irrigation and cultivation method on GWP of rice.

traits		GWP per unit area	GWP per unit weight	GWP per unit input energy	GWP per unit output energy
year	2019	1896^a	126.6^a	60^a	9.4^a
	2020	1580^b	102.4^b	56.9^b	7.7^b
Cultivation method	Direct-seeded cultivation	1681^b	124.1^a	59.2^a	9.4^a
	Un-puddled transplanted	1593^c	104.8^c	56.3^b	7.8^c
	Puddled transplanted	1940^a	114.7^b	59.8^a	8.5^b
Irrigation	Sprinkler Irrigation	1608^b	120.3^a	55^b	9.1^a
	Permanent flood	1817^a	112^b	59.4^a	8.3^b
	Periodic flood	1776^a	113.7^b	60.4^a	8.5^b
	Drip irrigation	1752^a	111.9^b	59^a	8.4^b

Means with the similar letters in each column are not significantly different at 5 % of probability level (LSD Test).

that the lowest amount of water consumption was obtained under the drip and sprinkler irrigation system. Also, the transplanted cultivation method had a higher yield than the direct-seeded cultivation method. This study showed that under different irrigation and cultivation scenarios, drip irrigation with un-puddled transplanted cultivation can be a technology to save water (46 % and 44 % saving of water consumption for the first and second year, respectively, compared to permanent flood irrigation with puddled transplanted cultivation) with yield preservation for areas with water crisis. Based on the results of mean comparison, the energy input in un-puddled transplanted and direct-seeded cultivation was lower than puddled transplanted. Also, the highest energy input and global warming potential in different irrigation systems were related to the permanent flood irrigation system, and in different cultivation methods, were related to the puddled transplanted. The results of mean comparison of cultivation method showed that there were no significant differences between the amounts of EUE, EP and SE for the puddled and un-puddled transplanted treatments. Also, the lowest amounts of EUE, EP and NE were obtained for the direct-seeded cultivation. According to the results of mean comparison of irrigation method, there were no significant differences between the amounts of EUE, EP, SE and NE for the periodic and permanent flood irrigation systems. The highest share of energy consumption in rice cultivation was related to irrigation water, nitrogen fertilizer and fuel, followed by the largest share of GWP related to the use of nitrogen fertilizers, fuels and machinery. So, it is suggested that future policies, should be inclined to promote biofuels instead of fossil fuels to reduces fuel energy for areas with energy crisis. The results of mean comparison of GWP per unit area, weight, input energy and output energy showed that there were no significant differences between drip, permanent and periodic flood irrigation systems. Also, the lowest amount of GWP per unit area, weight, input energy and output energy were related to the un-puddled transplanted treatment. The results of this study showed that un-puddled transplanted cultivation method (due to less energy input and a yield close to PTR) can be a suitable method for rice cultivation to reduce energy consumption and global warming potential.

Nomenclature

DSR	Direct-seeded rice	α	Coefficient of indirect irrigation energy
PTR	Puddled transplanted rice	W_c	Amount of water consumption
UPTR	Un-puddled transplanted rice	K_c	Rice plant coefficient
E_{in}	Energy input	CV	Coefficient of Variation
E_{out}	Energy output	LSD	Least significant difference
LCC	Leaf Color Chart for N application	df	Degree of Freedom
ET_0	Reference evapotranspiration	EUE	Energy use efficiency
K_p	Coefficient of the pan	EP	Energy productivity
E_{pan}	Pan evaporation	SE	Specific energy
ET_c	Rice plant evapotranspiration	NE	Net energy
E_{1a}	Energy of labor per hectare	GWP	Global warming potential
EI_{1a}	Equivalent energy of labor	CO_2	Carbon Dioxide
E_w	Energy of irrigation	GHG	Greenhouse gas
T	Working hours of the labor per day	N_2O	Nitrous oxide
N	Number of labor	CH_4	Methane

CRediT authorship contribution statement

Seyede Tayyebeh Hosseini: Writing – original draft, Resources, Methodology, Investigation, Data curation, Conceptualization. **Hossein Sharifan:** Writing – review & editing. **Alireza Kiani:** Writing – review & editing, Funding acquisition. **Noormohammad Abyar:** Writing – review & editing. **Mohammad taghi Feyzbakhsh:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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